The `luastats` Package in LaTeX

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1 Introduction

The luastats package is a LuaLaTeX-based statistical computation library designed to provide essential statistical functions using Lua scripting within LaTeX. It aims to bridge the gap between LaTeX's typesetting capabilities and Lua's computational power, enabling users to perform statistical calculations directly within their LaTeX documents without relying on external tools or complex macros.

2 Motivation Behind luastats

Statistical analysis is a fundamental aspect of many academic and research fields, including mathematics, physics, economics, and social sciences. While LaTeX excels at formatting mathematical expressions and reports, its built-in calculation capabilities are limited. Traditionally, users have relied on tools like R, Python, or MATLAB for data analysis before manually integrating results into their LaTeX documents. The luastats package eliminates this extra step by allowing users to compute statistical measures dynamically within their LaTeX workflow using Lua.

2.1 Key Features

1. Seamless Lua Integration: Leverages LuaLaTeX's embedded Lua scripting environment to perform calculations efficiently.
2. Essential Statistical Functions: Provides core statistical operations,including:
   - Mean (average) computation
   - Median calculation
   - Rounding functions
3. Dynamic Computation in LaTeX: Users can pass data directly in LaTeX and receive computed results in real time, enhancing automation in document generation.
4. User-Friendly Syntax: The package is designed to be intuitive, requiring minimal coding knowledge to use its functions.
5. Efficient Execution: Lua's lightweight nature ensures fast computations, making it a practical alternative to heavier computational tools.

3 Who Should Use?

- Researchers & Academics: Ideal for those who work with data-driven reports and need real-time statistical processing in LaTeX.
- Students & Educators: Helps in teaching statistics, probability, and mathematical analysis without switching between different software.
- Technical Writers: Useful for professionals preparing reports, articles, or books that require embedded calculations.

4 Installation and License

To install the `luastats` package, place the `luastats.sty` file in your LaTeX directory. Include the package in your document with:

```
\usepackage{luastats}
```

5 Basic Statistical Method

Code	**Uses**	**Output**
`\luaMean{1,2,3,4,5}`	Computes the arithmetic mean	3.0
`\luaMedian{1,2,3,4,5}`	Finds the median	3
`\luaMode{1,2,2,3,3,3,4,4}`	Determines mode	3
`\luaVariance{1,2,3,4,5}`	Computes variance	2.0
`\luaSD{1,2,3,4,5}`	Standard deviation	1.41421
`\luaRange{1,2,3,4,5}`	Range (max - min)	4
`\luaQuartiles{1,2,3,4,5,6,7,8,9,10}`	Computes quartiles	Q1: 3 , Q2: 5.5 , Q3: 8
`\luaIQR{1,2,3,4,5,6,7,8,9,10}`	Interquartile range	5
`\luaSkewness{1,2,3,4,5,6,7,8,9,10}`	Measures skewness	-6.167905692362e-17
`\luaKurtosis{1,2,3,4,5,6,7,8,9,10}`	Measures kurtosis	-0.46361719649871
`\luaCorrelation{1,2,3,4}{2,4,6,8}`	Pearson correlation	1.0
`\luaCovariance{1,2,3,4}{2,4,6,8}`	Covariance	3.33333
`\luaRegression{1,2,3,4}{2,4,6,8}`	Linear regression	Slope: 2.0 , Intercept: 0.0
`\luaChiSquare{5,10,15}{6,9,15}`	Chi-square test	0.27778
`\luaEMA{1,2,3,4,5}{0.5}`	Exponential moving average	4.0625
`\luaWeightedMean{1,2,3,4}{0.1,0.2,0.3,0.4}`	Weighted mean	3.0
`\luaZScores{1,2,3,4,5}`	Converts to Z-scores	-1.22474, 0.0, 1.22474

Table 1: Statistical Functions Reference

`\luaCSVsummary{data.csv}` Summary from CSV :

Column	Mean	Median	Mode	Variance	Std Dev	Range
1	5.50	5.50	1, 4, 10, 7	11.25	3.35	9.00
2	6.50	6.50	8, 5, 2, 11	11.25	3.35	9.00
3	7.50	7.50	12, 9, 6, 3	11.25	3.35	9.00
4	4.75	5.50	7	6.19	2.49	6.00
5	5.75	6.50	8	6.19	2.49	6.00
6	6.75	7.50	9	6.19	2.49	6.00
7	7.00	7.00	5, 9, 7	2.67	1.63	4.00

Code	Uses	Output
`\DataSet{data1}{1,2,3,4,5}`	Defines dataset	–
`\luastat{data1}{mean}`	Computes mean	5.5
`\luastat{data1}{median}`	Computes median	5.5
`\luastat{data1}{mode}`	Computes mode	1, 2, 3, 4, 5, 6, 7, 8, 9, 10
`\luastat{data1}{variance}`	Computes variance	8.25
`\luastat{data1}{standard_deviation}`	Computes standard deviation	2.87228
`\luastat{data1}{range}`	Computes range	9.0
`\luastat{data1}{quartiles}`	Computes quartiles	Q1: 3.00000, Q2: 5.50000, Q3: 8.00000
`\luastat{data1}{iqr}`	Computes interquartile range	5.0
`\luastat{data1}{skewness}`	Measures skewness	0.0
`\luastat{data1}{kurtosis}`	Measures kurtosis	-0.46362

Table 2: Statistical Functions for a Single Dataset

The package provides a comprehensive set of statistical functions for LaTeX documents. It simplifies the inclusion of PMFs, PDFs, and CDFs for various distributions, making it a valuable tool for researchers, statisticians, and students.

- Probability Mass Functions (PMFs)
- Probability Density Functions (PDFs)
- Cumulative Distribution Functions (CDFs)

Name and Syntax	**Output**
Binomial `\luabinomialPMF`	$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}, \quad k=0,1,\ldots,n$
Geometric `\luageometricPMF`	$P(X=k)=(1-p)^{k-1}p, \quad k=1,2,3,\ldots$
Negative Binomial `\luanegBinomialPMF`	$P(X=k)=\binom{k-1}{r-1}p^r(1-p)^{k-r}, \quad k=r,r+1,\ldots$
Poisson `\luapoissonPMF`	$P(X=k)=\frac{\lambda^k e^{-\lambda}}{k!}, \quad k=0,1,2,\ldots$
Hypergeometric `\luahypergeometricPMF`	$P(X=k)=\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}, \quad \max(0,n-(N-K))\leq k\leq \min(n,K)$
Discrete Uniform `\luadiscreteUniformPMF`	$P(X=x)=\frac{1}{n}, \quad x=1,2,\ldots,n$
Multinomial `\luamultinomialPMF`	$P(X_1=k_1,\ldots,X_m=k_m)=\frac{n!}{k_1!k_2!\ldots k_m!}p_1^{k_1}p_2^{k_2}\ldots p_m^{k_m}$
Zipf `\luazipfPMF`	$P(X=k)=\frac{k^{-\alpha}}{\zeta(\alpha)}, \quad k=1,2,3,\ldots$
Negative Hypergeometric `\luanegHypergeometricPMF`	$P(X=k)=\frac{\binom{k-1}{r-1}\binom{N-k}{K-r}}{\binom{N}{K}}, \quad k=r,r+1,\ldots,N-K+r$

Table 3: Probability Mass Functions (PMFs)

6 Plotting Probability Distributions

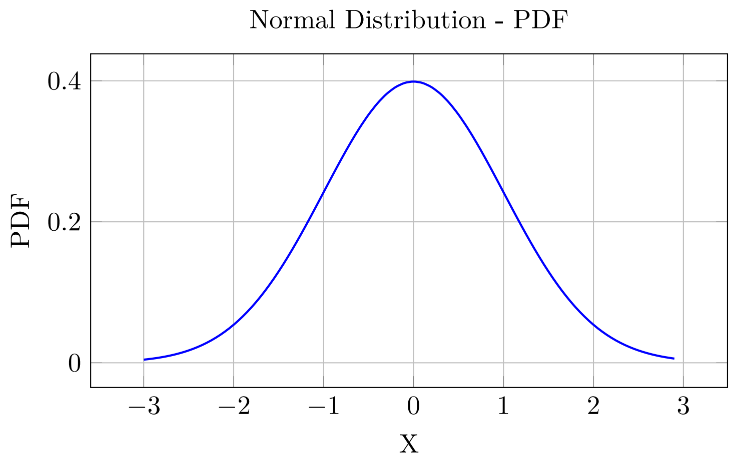
This package provides plotting functionality for different distributions using `TikZ` and `pgfplots`.

6.1 Normal Distribution

6.1.1 Probability Density Function (PDF)

```
\luanormpdf{mean}{stddev}{range}{step}
```

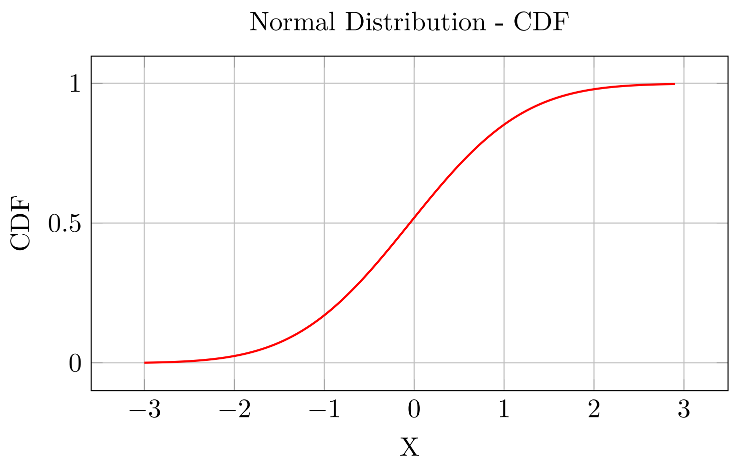
Example:



6.1.2 Cumulative Distribution Function (CDF)

```
\luanormcdf{mean}{stddev}{range}{step}
```

Example:

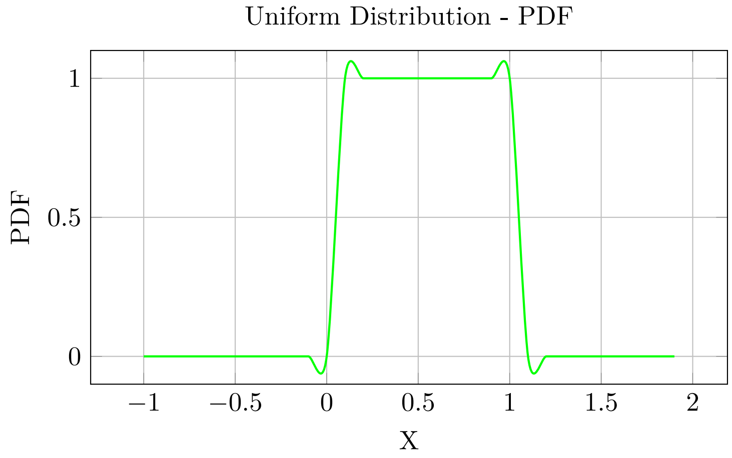


6.2 Uniform Distribution

6.2.1 Probability Density Function (PDF)

```
\luaunipdf{a}{b}{range}
```

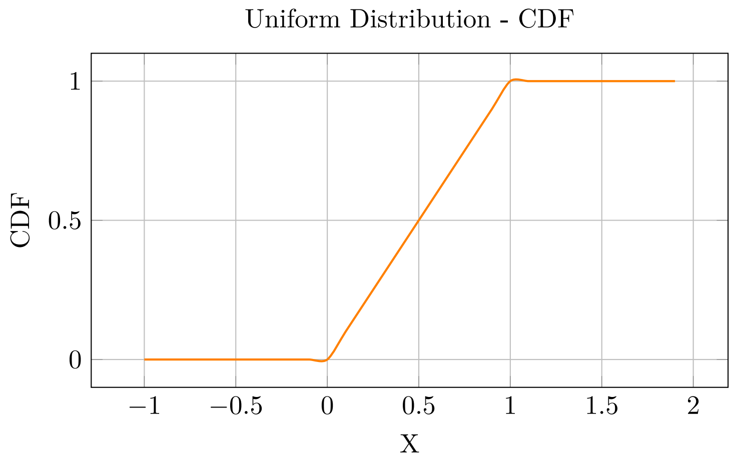
Example:



6.2.2 Cumulative Distribution Function (CDF)

```
\luaunicdf{a}{b}{range}
```

Example:

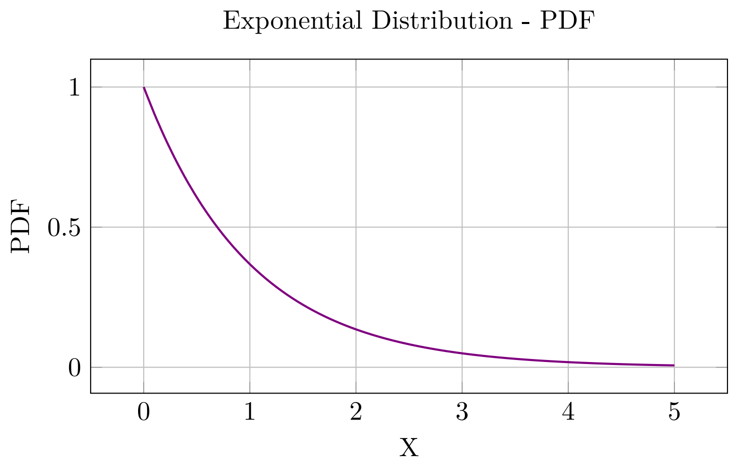


6.3 Exponential Distribution

6.3.1 Probability Density Function (PDF)

```
\luaexppdf{lambda}{range}{step}
```

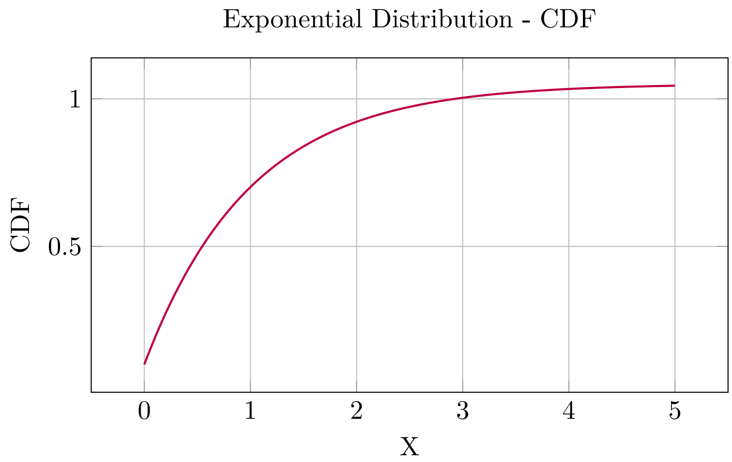
Example:



6.3.2 Cumulative Distribution Function (CDF)

```
\luaexpcdf{lambda}{range}{step}
```

Example:

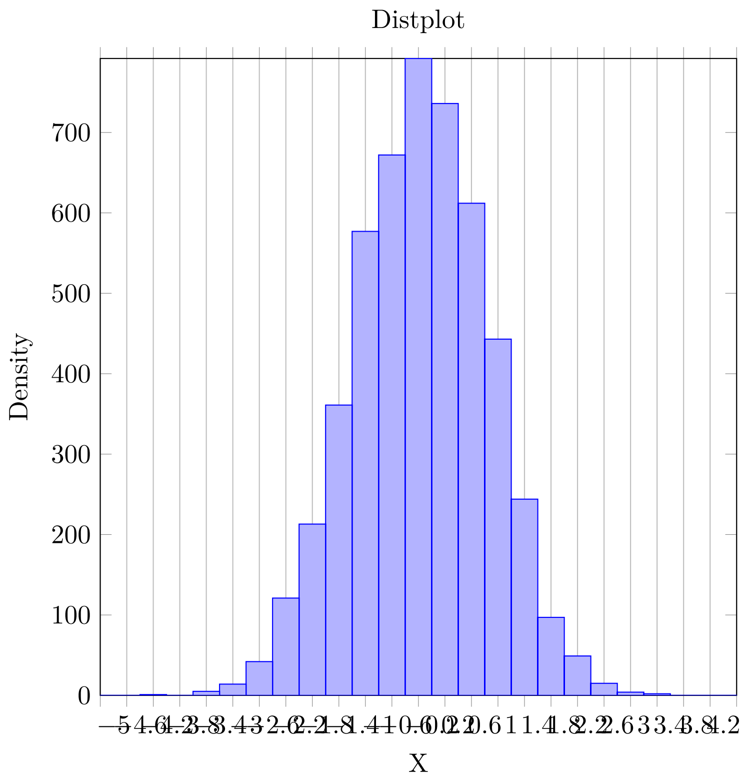


7 Histogram and KDE (Kernel Density Estimation)

7.1 Histogram and Smoothed KDE

```
\luadistplot{data}{bins}{range}{step}
```

Example:



Name	Syntax	Output
Uniform Distribution	`\luauniPDF`	$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{otherwise} \end{cases}$
Normal (Gaussian) Distribution	`\luanormalPDF`	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
Exponential Distribution	`\luaexpPDF`	$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$
Gamma Distribution	`\luagammaPDF`	$f(x) = \begin{cases} \frac{\lambda^k x^{k-1} e^{-\lambda x}}{\Gamma(k)}, & x > 0 \\ 0, & x \leq 0 \end{cases}$
Beta Distribution	`\luabetaPDF`	$f(x) = \begin{cases} \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$
Cauchy Distribution	`\luacauchyPDF`	$f(x) = \frac{1}{\pi\gamma} \left[1 + \left(\frac{x-x_0}{\gamma} \right)^2 \right]^{-1}$
Laplace Distribution	`\lualaplacePDF`	$f(x) = \frac{1}{2b} e^{-\frac{\lvert x-\mu \rvert}{b}}$
Student's t Distribution	`\luastudentTPDF`	$f(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}$
Weibull Distribution	`\luaweibullPDF`	$f(x) = \begin{cases} \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}, & x \geq 0 \\ 0, & x < 0 \end{cases}$
Pareto Distribution	`\luaparetoPDF`	$f(x) = \begin{cases} \frac{\alpha x_m^\alpha}{x^{\alpha+1}}, & x \geq x_m \\ 0, & x < x_m \end{cases}$
Rayleigh Distribution	`\luarayleighPDF`	$f(x) = \begin{cases} \frac{x}{\sigma^2} e^{-\frac{x^2}{2\sigma^2}}, & x \geq 0 \\ 0, & x < 0 \end{cases}$
Log-Normal Distribution	`\lualognormalPDF`	$f(x) = \begin{cases} \frac{1}{x\sigma\sqrt{2\pi}} e^{-\frac{(\ln x-\mu)^2}{2\sigma^2}}, & x > 0 \\ 0, & x \leq 0 \end{cases}$

Table 4: Probability Density Functions (PDFs)

Name	Syntax	Output
Uniform Distribution	`\luauniCDF`	$F(x) = \begin{cases} 0, & x < a \\ \frac{x-a}{b-a}, & a \le x \le b \\ 1, & x > b \end{cases}$
Normal (Gaussian) Distribution	`\luanormalCDF`	$F(x) = \frac{1}{2}\left[1 + \text{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right]$
Exponential Distribution	`\luaexpCDF`	$F(x) = \begin{cases} 1 - e^{-\lambda x}, & x \ge 0 \\ 0, & x < 0 \end{cases}$
Gamma Distribution	`\luagammaCDF`	$F(x) = \begin{cases} \frac{\gamma(k,\lambda x)}{\Gamma(k)}, & x > 0 \\ 0, & x \le 0 \end{cases}$
Beta Distribution	`\luabetaCDF`	$F(x) = \begin{cases} I_x(\alpha, \beta), & 0 \le x \le 1 \\ 0, & x < 0 \\ 1, & x > 1 \end{cases}$
Cauchy Distribution	`\luacauchyCDF`	$F(x) = \frac{1}{\pi}\tan^{-1}\left(\frac{x-x_0}{\gamma}\right) + \frac{1}{2}$
Laplace Distribution	`\lualaplaceCDF`	$F(x) = \begin{cases} \frac{1}{2}e^{\frac{x-\mu}{b}}, & x < \mu \\ 1 - \frac{1}{2}e^{-\frac{x-\mu}{b}}, & x \ge \mu \end{cases}$
Student's t Distribution	`\luastudentTCDF`	$F(x) = I_t(\frac{\nu}{2}, \frac{1}{2})$
Weibull Distribution	`\luaweibullCDF`	$F(x) = \begin{cases} 1 - e^{-(x/\lambda)^k}, & x \ge 0 \\ 0, & x < 0 \end{cases}$
Pareto Distribution	`\luaparetoCDF`	$F(x) = \begin{cases} 1 - \left(\frac{x_m}{x}\right)^{\alpha}, & x \ge x_m \\ 0, & x < x_m \end{cases}$
Rayleigh Distribution	`\luarayleighCDF`	$F(x) = \begin{cases} 1 - e^{-\frac{x^2}{2\sigma^2}}, & x \ge 0 \\ 0, & x < 0 \end{cases}$
Log-Normal Distribution	`\lualognormalCDF`	$F(x) = \begin{cases} \frac{1}{2}\left[1 + \text{erf}\left(\frac{\ln x-\mu}{\sigma\sqrt{2}}\right)\right], & x > 0 \\ 0, & x \le 0 \end{cases}$

Table 5: Cumulative Distribution Functions (CDFs)

Distribution	**Parameters**	**Generated Values**
Normal	$\mu = 0, \sigma = 1, n = 5$ `\luarandomnormal{0}{1}{5}`	-1.6808, 1.8769, 0.5560, 0.6659, -0.2570,
Uniform	$a = 0, b = 1, n = 5$ `\luarandomuniform{0}{1}{5}`	0.7736, 0.8342, 0.5923, 0.0584, 0.2588,
Exponential	$\lambda = 1, n = 5$ `\luarandomexponential{1}{5}`	2.3260, 0.9571, 2.3697, 0.1581, 0.4720,
Binomial	$n = 10, p = 0.5, samples = 5$ `\luarandombinomial{10}{0.5}{5}`	5.0000, 6.0000, 5.0000, 4.0000, 6.0000,
Poisson	$\lambda = 3, samples = 5$ `\luarandompoisson{3}{5}`	1.0000, 2.0000, 1.0000, 4.0000, 2.0000,
Chi-Square	$df = 5, samples = 5$ `\luarandomchisquare{5}{5}`	4.5152, 1.4233, 8.1190, 14.8067, 1.3692,
t-Distribution	$df = 10, samples = 5$ `\luarandomt{10}{5}`	-0.5001, 0.1082, -2.4687, -0.4851, -0.9979,
F-Distribution	$df_1 = 5, df_2 = 10, samples = 5$ `\luarandomf{5}{10}{5}`	1.5766, 0.6876, 0.1622, 1.0725, 0.4508,

Table 6: Random Number Generation from Various Distributions